Hailay Kidu Teklehaymanot

Contact Information

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Professional Summary

Ph.D. candidate at **L3S Research Center, Leibniz Universität Hannover** (Supervisor: Prof. Wolfgang Nejdl), with expected defense in Summer 2026. DAAD scholarship recipient with over 10 years of academic experience teaching machine Learning and NLP at Ethiopian Universities, supervising graduate students, and leading research projects. Expertise in Multilingual and Low-resource NLP as well as Retrieval-Augmented Generation (RAG). Skilled in grant proposal development for programs such as NUFFIC, Erasmus+, EU Horizon, and Shimedit Science, with experience in consortium coordination, international collaborations, and interdisciplinary project leadership. Entrepreneur and founder of SMH Electronics and Engineering, a Technology startup in Ethiopia. Winner of the SIGIR 2026 LiveRAG Challenge and the AI Grid Hackathon 2024.

Professional Experience

2022–Present **PhD Researcher**, L3S Research Center, Leibniz Universität Hannover, Germany
Research on **Linguistically Grounded NLP for Low-resource Languages**; developed Morphology-Aware Subword Construction (MoVoC) and Lexically Grounded Subword Embeddings (LGSE); designed RAG pipelines (OpenSearch, Pinecone); coordinated EU Horizon and Shimedit Science proposals; international collaborations with University of Zurich, and Mekelle University

2010–2019 **Assistant Professor**, Mekelle University & University of Gondar, Ethiopia
Led AI/NLP research groups; supervised MSc theses; developed AI curricula; initiated research labs; promoted international collaborations

2007–2009 **Data Scientist**, Cyber Software, Ethiopia
Built ML models for National Justice Information System; developed analytics dashboards and reporting pipelines

2007–2009 **Early Doctoral Researcher**, Beijing University of Technology, China
Conducted preliminary research in NLP ; recipient of the **China Scholarship Council (CSC) Scholarship**

Education

2022–Present **PhD in Computer Science**, L3S Research Center, Leibniz Universität Hannover (Supervisor: Prof. Wolfgang Nejdl)
DAAD Scholarship, SIGIR 2026 LiveRAG Challenge Winner, AI Grid Hackathon 2024

2010–2012 **MSc in Information Technology**, University of Madras, India

2004–2007 **B.Ed. in Information Technology**, Mekelle University, Ethiopia

Skills & Competencies

AI / Machine Learning	Retrieval-Augmented Generation (RAG), Transformer Models, Representation Learning, Text Embeddings (dense/sparse), Sentence Transformers, Cross-lingual Embeddings, PyTorch, TensorFlow, Scikit-learn, MLflow
NLP & LLMs	Multilingual NLP, Low-Resource Language Processing, Tokenization (BPE, Morpheme-aware), Machine Translation, Question Answering, Information Retrieval, Embedding-based Retrieval, Reranking (BGE), Prompt Engineering
Programming & Systems	Python, MATLAB, Bash, R, SQL, NoSQL, Linux/UNIX, High-Performance Computing (HPC), Distributed Training
Research & Grants	EU Horizon proposals, Schmidt Science Fellowships applications, grant writing, consortium coordination, Git/GitLab, Agile workflows
Languages	Amharic (Native), Tigrinya (Native), English (Fluent), German (B1)

Selected Publications

Full list: Google Scholar (14 papers)

1. MoVoC: Morphology-Aware Subword Construction for Ge'ez Script Languages, Findings of Empirical Methods in Natural Language Processing(EMNLP) 2025
2. LGSE: Lexically Grounded Subword Embedding Initialization for Low-Resource Language Adaptation Language Resources and Evaluation Conferenc(LREC) 2026
3. TIGQA: An Expert Annotated Question Answering Dataset in Tigrinya, Language Resources and Evaluation Conference(LREC-COLING) 2024
4. Morphological Synthesizer for Ge'ez Language: Addressing Morphological Complexity and Resource Limitations, RAIL 2024
5. Tokenization Disparities as Infrastructure Bias: How Subword Systems Create Inequities in Large Language Model Access and Efficiency, FLLM Conference, IEEE Xplore 2025
6. RAGtifier: Evaluating Retrieval-Augmented Generation Approaches of State-of-the-Art RAG Systems for the SIGIR LiveRAG Competition, 2025
7. Low-Resource English–Tigrinya Machine Translation: Leveraging Multilingual Models, Custom Tokenizers, and Clean Evaluation Benchmarks, FLLM Conference, IEEE Xplore 2025
8. An Effective Sarcasm Detection Approach Based on Sentimental Context and Individual Expression Habits, Cognitive Computation, 2022
9. User Response-Based Fake News Detection on Social Media, ICAI 2021
10. A Mobile-Based Tigrigna Language Learning Tool, IAOE 2015

Awards & Recognition

2026	SIGIR LiveRAG Challenge Winner
2025	AI Grid Hackathon Recognition
2022	DAAD Scholarship (Germany)
2019	China Scholarship Council (CSC) Scholarship

References

Prof. Wolfgang Nejdl	Director, L3S Research Center | Professor, Leibniz Universität Hannover nejdl@l3s.de | www.l3s.de
Dr. Simon Gottschalk	Research Group Leader, L3S Research Center gottschalk@l3s.de | +49 511 762-17739
Dr. Daniel Kudenko	Research Group Leader, L3S Research Center kudenko@l3s.de | +49 511 762-17775